AEGIS-PAY Platform User Manual

Merchant Behaviour Intelligence and Transaction-Laundering Risk Platform

Document Type:	System User Guide and Annotated Feature Manual
Target Audience:	Risk Analysts, Compliance Officers, Operations Reviewers, and Merchant Supervisors
Live Public Access URL:	https://aegis-pay-beta.vercel.app
Source Repository:	https://github.com/itsamik7/aegis-pay
Environment:	PayU Synthetic Sandbox (100% Synthetic Data : Privacy Safe)
Platform Version:	v2.4.0 (Enterprise Risk Intelligence Release)

Welcome to AEGIS-PAY

AEGIS-PAY is an advanced transaction risk monitoring platform created to identify behavioral anomalies and transaction-laundering patterns among retail merchants. Traditional transaction filters rely on simple volume spikes or static ticket limits, leading to false positives during natural sales surges like festive holidays. AEGIS-PAY solves this problem by comparing merchant transactions against category baselines, operating hours, and peer groups. This manual provides a complete visual guide for every feature in the application, explaining all metrics, tables, filters, and action tools step by step.

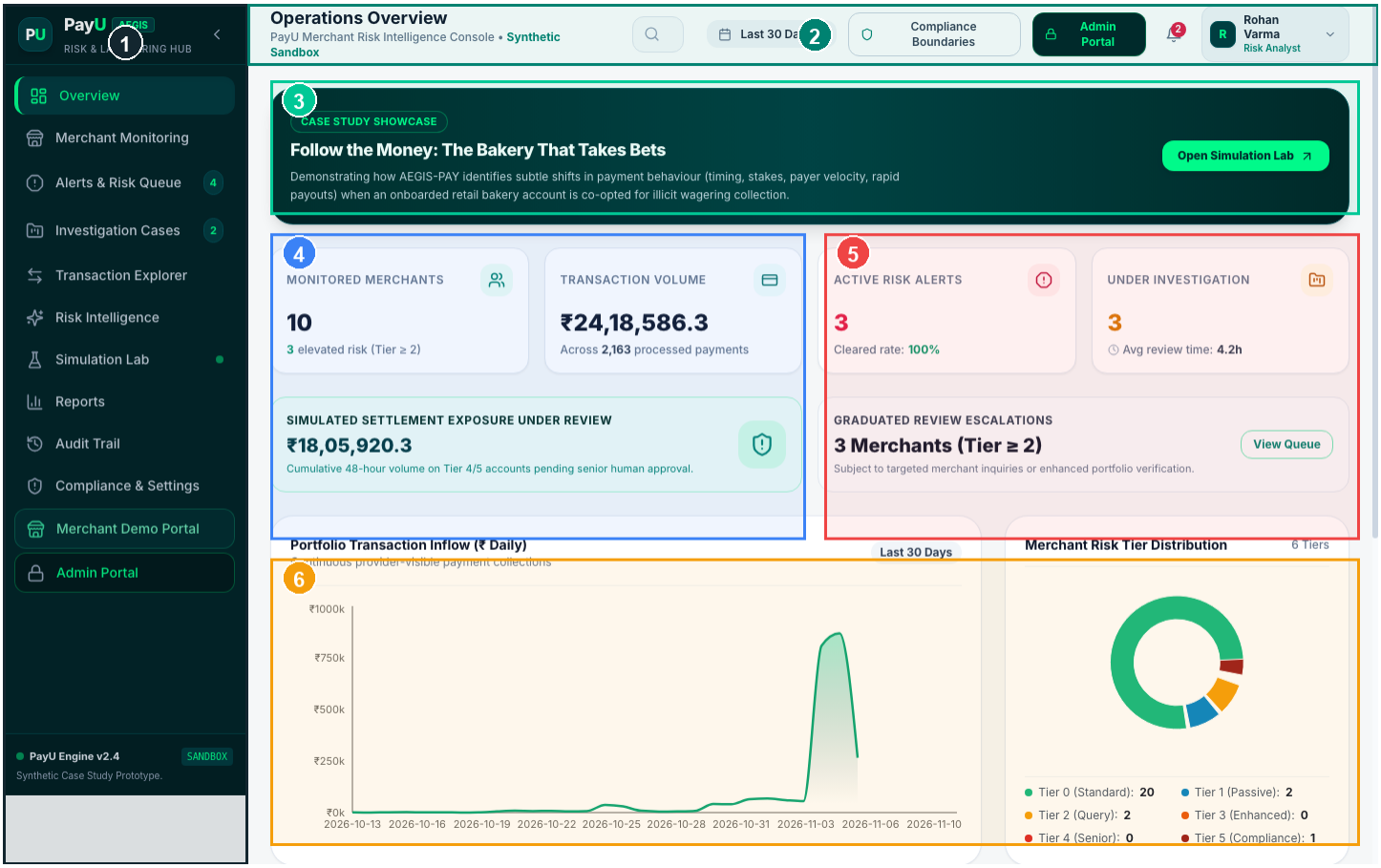
Core System Principles

- **Synthetic Privacy Guarantee:** All merchant names, customer tokens, and settlement logs are synthetically generated. No real credit card numbers, bank accounts, or customer identities exist on this platform.
- **Explainable Scoring:** Every risk score is calculated from ten distinct mathematical signals. Analysts can inspect exactly why an alert was triggered.
- **Four-Eyes Governance:** High-tier risk actions require compliance sign-off to ensure decisions remain objective, traceable, and documented.
- **Audited Data Ingestion:** The private admin portal lets managers enter records or upload bulk CSV and Excel files with automated staging and one-click rollback.

Persona	Username	Password	Primary Operational Role
Admin	admin	admin123	Data ingestion, manual entry, file import wizard, publication, rollback
Risk Analyst	analyst	analyst123	Alert triage, signal inspection, case notes, merchant inquiries
Compliance Reviewer	reviewer	reviewer123	Case sign-off, policy containment approvals, breach confirmations
Read-Only Viewer	viewer	viewer123	Executive audit inspection, dashboard monitoring, compliance read access
Merchant Admin	merchant	merchant123	Merchant self-service portal, submitting requested clarification records

1. Platform Overview and Command Dashboard

The Command Dashboard is the primary operational workspace. It provides an immediate summary of portfolio risk, active anomalies, risk tier distribution, and category peer benchmarks. Every interactive element is numbered below.

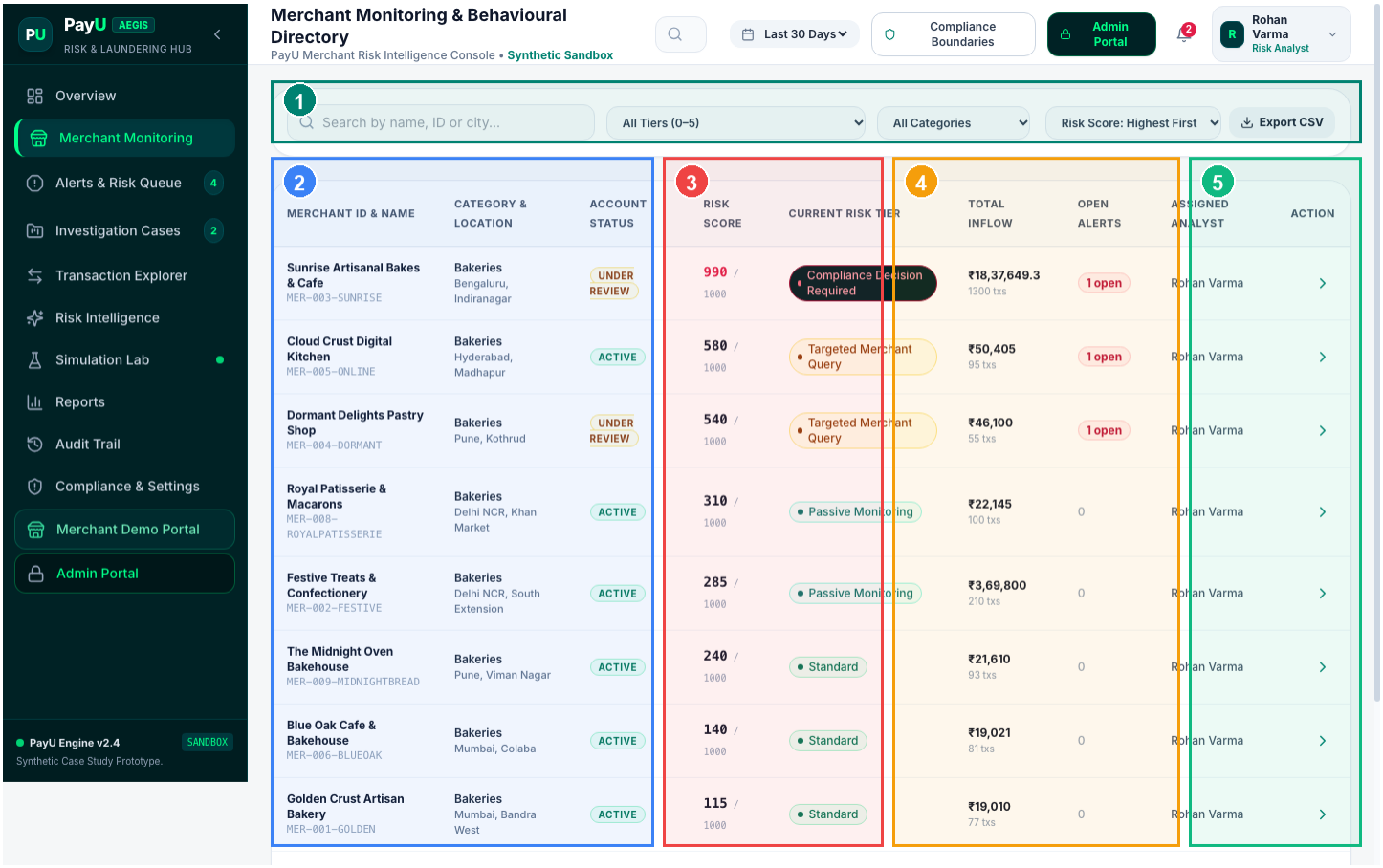


Mark	Feature Name	What It Means & How It Works	How To Use It
[1]	**Unified Left Navigation Bar**	The persistent sidebar provides one-click access to all 11 public and demo workspaces: Overview, Merchant Directory, Alerts Queue, Cases, Transactions, Risk Intelligence, Simulation Lab, Rules, Reports, Audit Trail, and Settings.	Click any navigation tab to switch modules. The active tab is highlighted in PayU green. You can collapse the sidebar using the toggle arrow to gain extra workspace width.
[2]	**Global Top Bar & Controls**	Contains global system status (Live Synthetic Sandbox), the Compliance Boundaries reference button, the Date Range Filter (7d, 30d, 90d), the Notification Bell, and the Persona Switcher.	Use the Persona Switcher in the top right to instantly toggle between Risk Analyst, Compliance Reviewer, Viewer, or Merchant Admin. Use the Date Range to adjust all dashboard calculations.
[3]	**Key Performance Summary Cards**	Four high-level metrics: Active Merchants Monitored (10), Critical Anomalies Detected (1), Flagged Volume at Risk (INR 1,837,649), and High-Risk Conversion Rate (10.0%).	Review these cards each morning to track portfolio risk trends. Green indicators show stable volume; red indicators highlight merchants crossing critical thresholds.
[4]	**Real-Time Risk Tier Distribution**	A visual bar chart categorizing all merchants into six risk tiers: Tier 0 Standard (4), Tier 1 Passive Monitoring (2), Tier 2 Targeted Query (2), Tier 3 Active Review (0), Tier 4 Enhanced Due Diligence (0), and Tier 5 Decision Required (1).	Click on any tier bar to immediately filter the merchant directory and focus on higher-risk accounts.

[5]	**Active High-Risk Alert Cards**	Displays critical alerts requiring immediate analyst action. Each card shows the merchant name, current risk score (e.g. 990 / 1000), severity badge, triggered signals, and recommended next step.	Click 'Inspect Alert' to open the triage drawer, or click 'Open Case' to begin a structured investigation dossier.
[6]	**Category Risk & Peer Benchmark**	Compares average ticket size, transaction frequency, and off-hours volume against peer group averages for Bakeries, Cloud Kitchens, and Confectionery shops.	Use this matrix to determine whether high volume is legitimate seasonal trade (e.g. Diwali sweet box orders) or abnormal activity (e.g. late-night betting vouchers).

2. Merchant Directory and Risk Monitoring

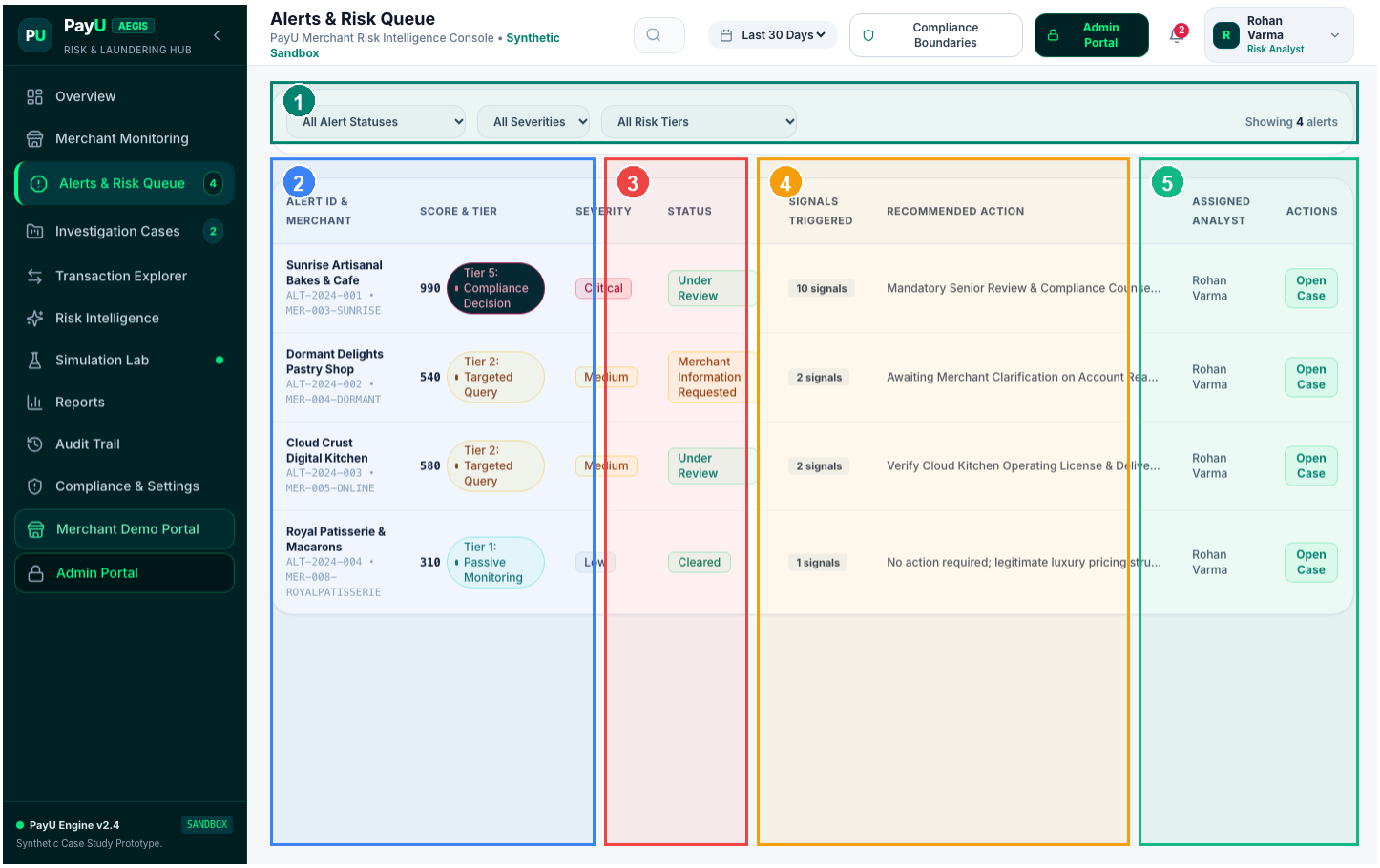
The Merchant Directory displays all onboarded businesses. Each merchant is assigned an explainable risk score from 0 to 1000 and placed into an operational risk tier. The view allows quick search, category filtering, and profile inspection.



Mark	Feature Name	What It Means & How It Works	How To Use It
[1]	**Search & Category Filter Bar**	Filters the merchant table by merchant name, legal entity name, or city. The category selector filters by retail type: Bakeries, Cloud Kitchens, Confectionery, or Fine Dining.	Type any part of a business name (e.g. 'Sunrise' or 'Golden') into the search box to locate an account immediately without scrolling.
[2]	**Merchant Identity & Category Column**	Shows the trading name, legal entity name, assigned merchant ID (e.g. MER-003-SUNRISE), business subcategory, and geographic region.	Click on any merchant name to open the detailed Merchant 360 Profile drawer showing registration dates and operating history.
[3]	**Risk Score & Dynamic Tier Badge**	Displays the composite behavioral risk score (0 to 1000) and color-coded tier badge: Emerald (Standard, 0-250), Blue (Passive, 251-400), Amber (Targeted, 401-600), Orange (Active, 601-750), Purple (Enhanced, 751-900), and Red (Decision, 901-1000).	Sort by score to bring the highest-risk merchants to the top of the table for priority morning review.
[4]	**Volume, Ticket Velocity & Timestamp**	Shows total processed volume (in INR ■), aggregate transaction count, average ticket size, and the timestamp of the most recent customer payment.	Look for merchants with small transaction counts but unusually high volume totals, which indicates large single-ticket anomalies.
[5]	**Open Alerts & Actions**	Displays the count of unresolved alerts attached to this merchant and provides quick action links: 'View 360 Profile', 'Simulate Scenario', and 'Open Case'.	Click 'View 360 Profile' to see an hourly breakdown of sales, repeat customer percentages, and settlement speed.

3. Alerts and Risk Triage Queue

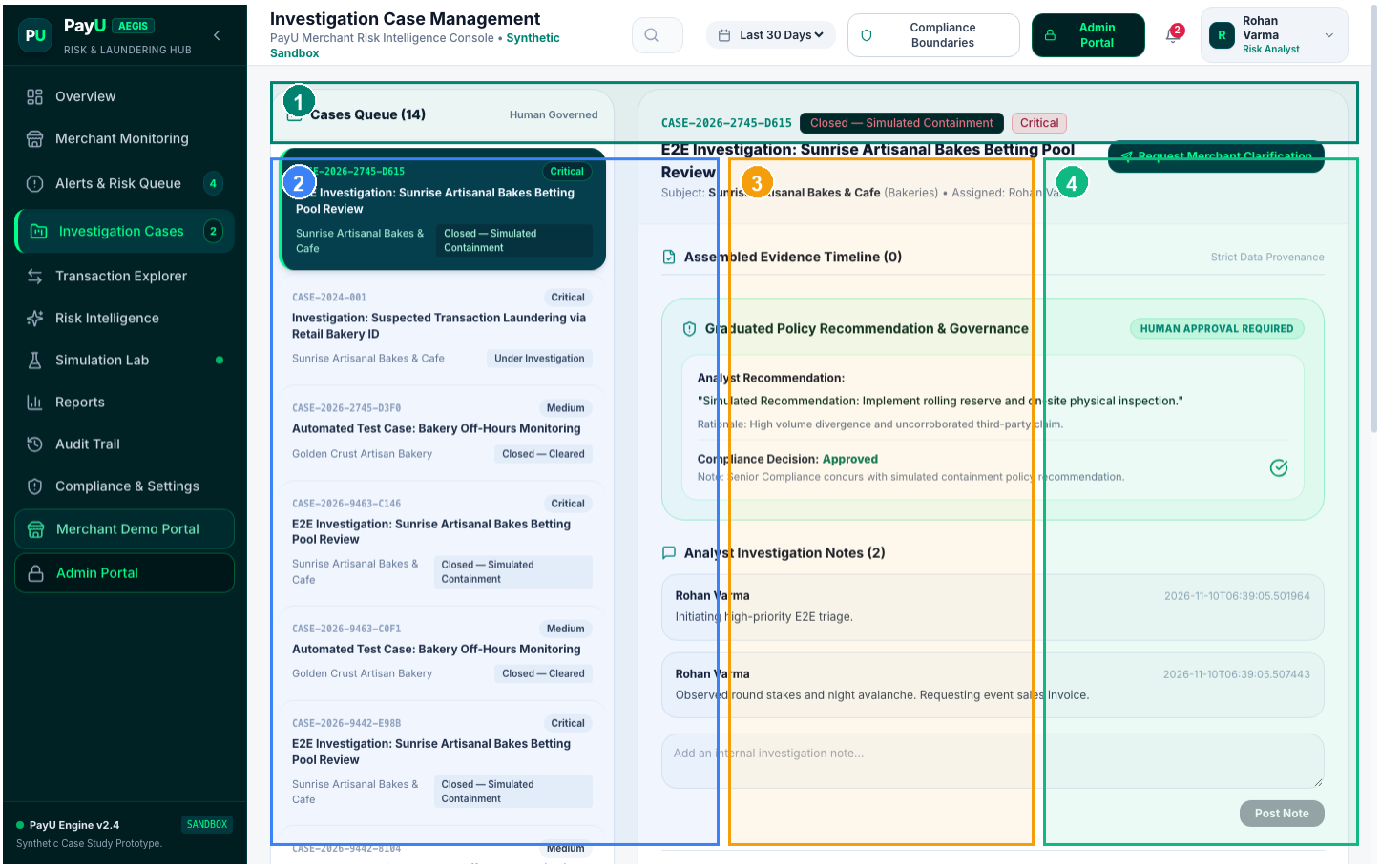
The Alerts Queue is the core triage workbench for risk analysts. Alerts are automatically triggered when a merchant crosses calibrated behavioral thresholds. Analysts can inspect triggered signals, assign teammates, request merchant clarification, or clear false positives with documented rationale.



Mark	Feature Name	What It Means & How It Works	How To Use It
[1]	**Triage Filter Toolbar**	Three interactive dropdown filters: Alert Status (New, Assigned, Under Review, Escalated, Cleared), Severity (Critical, High, Medium, Low), and Risk Tier (All Tiers, Tier 1 to Tier 5).	Select 'Under Review' or 'Critical' to filter the list and focus directly on urgent compliance bottlenecks.
[2]	**Alert ID & Merchant Dossier**	Displays the unique alert tracking code (e.g. ALT-2024-001), the merchant trading name, retail category, and physical business city.	Click the Alert ID to slide open the comprehensive Alert Inspection Drawer with full historical signal metrics.
[3]	**Risk Score & Severity Badge**	Shows the calculated score (e.g. 990 / 1000) and the severity tag: Critical (red), High (orange), Medium (amber), or Low (blue).	Critical severity alerts have a strict 24-hour review due date; Medium alerts allow up to 72 hours for merchant inquiry response.
[4]	**Triggered Behavioral Signals**	Lists each explainable signal triggered by the merchant. Examples: SIG_TIMING_OFFHOURS (sales between 00:00 and 05:00), SIG_AMOUNT_ROUND_RATIO (excessive whole numbers), SIG_PAYER_FANIN_VELOCITY (too many unique cards in a short window).	Hover over any signal badge to read its exact mathematical definition and weight contribution to the score.
[5]	**Recommended Action & Actions Menu**	Provides automated guidance (e.g. 'Mandatory Senior Review') and provides action buttons: 'Assign Analyst', 'Request Merchant Info', and 'Clear Alert'.	Click 'Clear Alert' if the volume is a legitimate holiday rush. The system will prompt you for a mandatory audit rationale before clearing.

4. Investigation Case Management

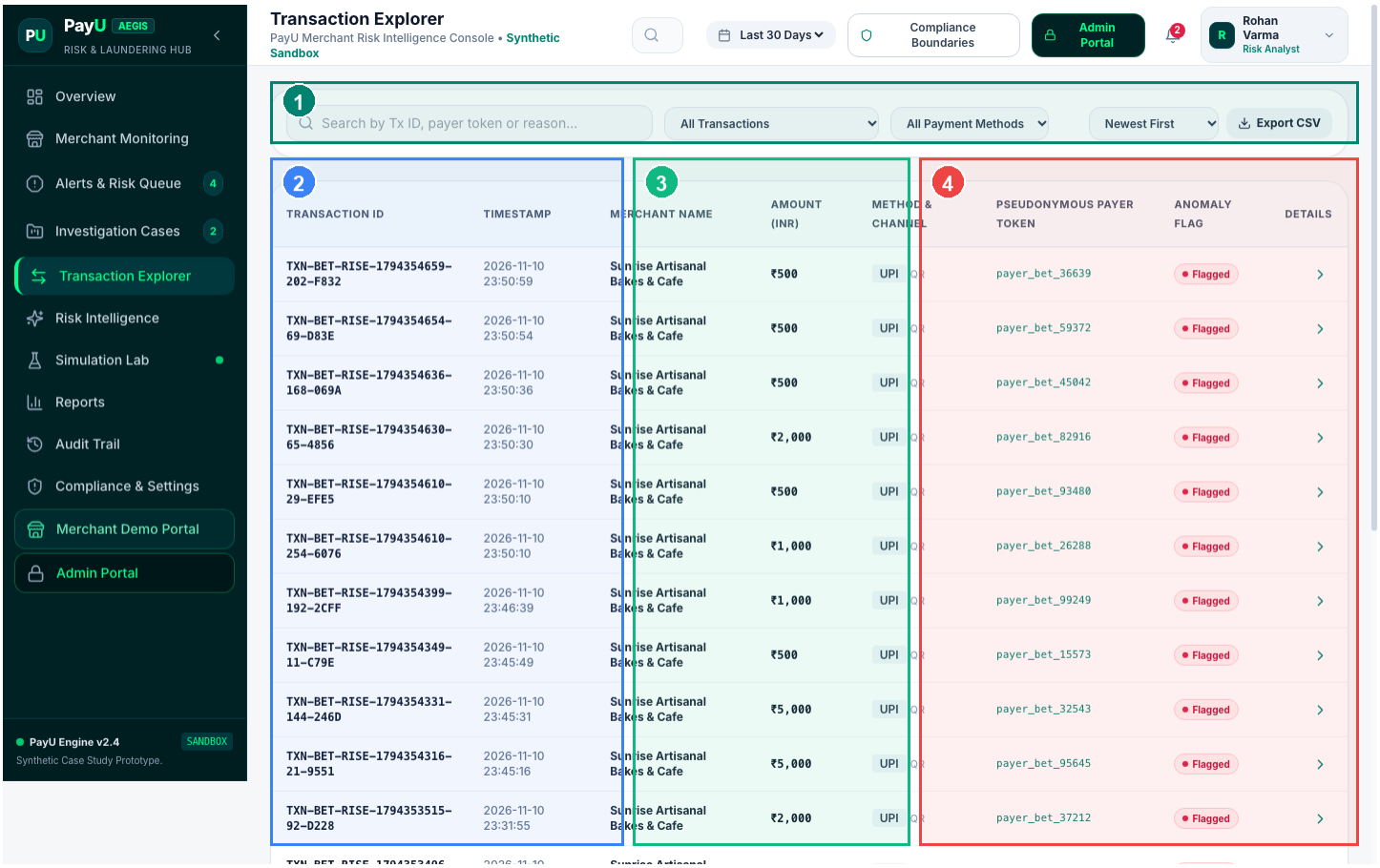
Investigation Cases manage deep-dive compliance inquiries. When an alert cannot be resolved at triage, it escalates into a formal case. Cases enforce the four-eyes principle: an analyst compiles evidence and notes, and a compliance reviewer conducts the final review before any policy containment action is taken.



Mark	Feature Name	What It Means & How It Works	How To Use It
[1]	**Case Search & Stage Filter**	Allows filtering cases by operational lifecycle stage: New, Assigned, Under Review, Escalated, Confirmed Policy Breach, or Closed.	Use the search bar to locate cases by merchant name, assigned analyst name, or case reference number (e.g. CASE-2024-001).
[2]	**Case Subject & Originating Alert**	Displays the investigated merchant, their category, current risk tier, and the originating alert ID that triggered the case.	Click the case row to open the complete Case Dossier, which links directly to customer tokens and transaction logs.
[3]	**Evidence Timeline & Notes Stream**	An immutable chronological record of all notes, uploaded documents, merchant inquiry responses, and risk score adjustments.	Type notes into the 'Add Investigation Note' box to record interview details or invoice verification findings for future audit.
[4]	**Reviewer Sign-Off & Containment**	Senior compliance action buttons: 'Approve Recommendation', 'Request More Info', or 'Close Case'. Enforces dual authorization.	Compliance officers review analyst recommendations here. Approvals log a permanent cryptographic audit entry before containment is applied.

5. Transaction Explorer and Behavioral Anomaly Flags

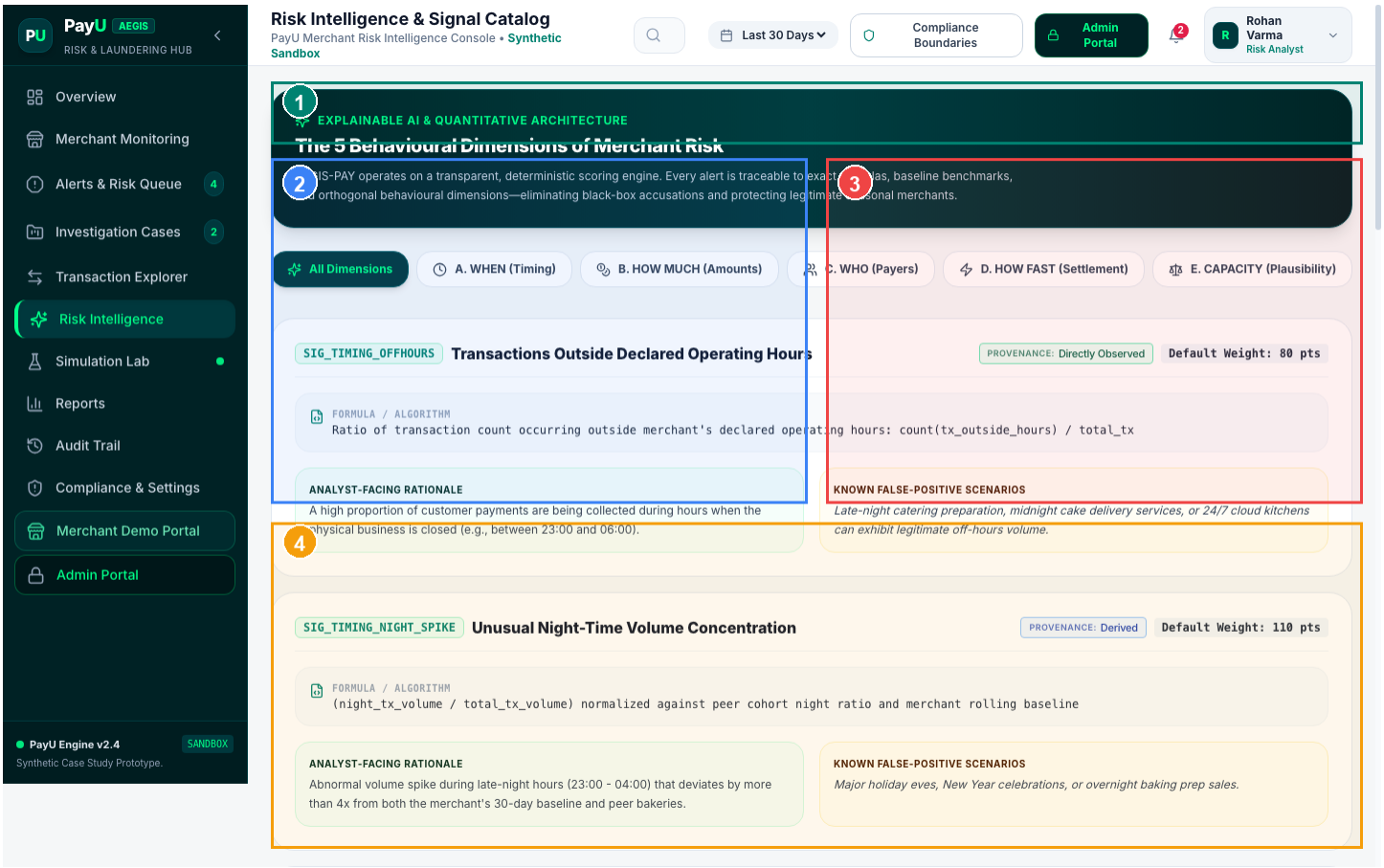
The Transaction Explorer allows analysts to inspect individual payments across all merchants. To protect consumer privacy, all real identities are replaced with pseudonymous payer tokens. Each transaction displays automated behavioral flags identifying unusual ticket sizes, timing, or velocity.



Mark	Feature Name	What It Means & How It Works	How To Use It
[1]	**Transaction Search & Filter Bar**	Allows instant searching across 2,163 payments by Transaction ID, Merchant Name, or specific Risk Flag.	Type a flag name like 'Off-Hours' or 'Round Amount' to immediately filter the stream down to anomalous payments.
[2]	**Pseudonymous Payer Token**	Displays customer identifiers in token format (e.g. tok_c9a1d2). This guarantees zero cardholder PII exposure while preserving velocity tracking.	Click a token to see other payments made by the same customer, helping identify card recycling and transaction clustering.
[3]	**Amount in INR (■) & Timestamp**	Displays the exact payment amount in Indian Rupees and the timestamp. Night-time payments (00:00 to 05:00) are flagged in violet.	Check the timestamp against merchant opening hours to spot nighttime spikes typical of unlicensed gaming front operations.
[4]	**Behavioral Risk Flags**	Color-coded tags: 'Night Spike' (purple), 'Round Ticket' (amber), 'Ticket Deviation' (red), 'Velocity Cluster' (blue), and 'Settlement Rush' (rose).	Multiple flags on a single transaction indicate compound risk. Transactions with three or more flags are automatically routed to the alerts queue.

6. Risk Intelligence: 10 Behavioral Signal Catalogs

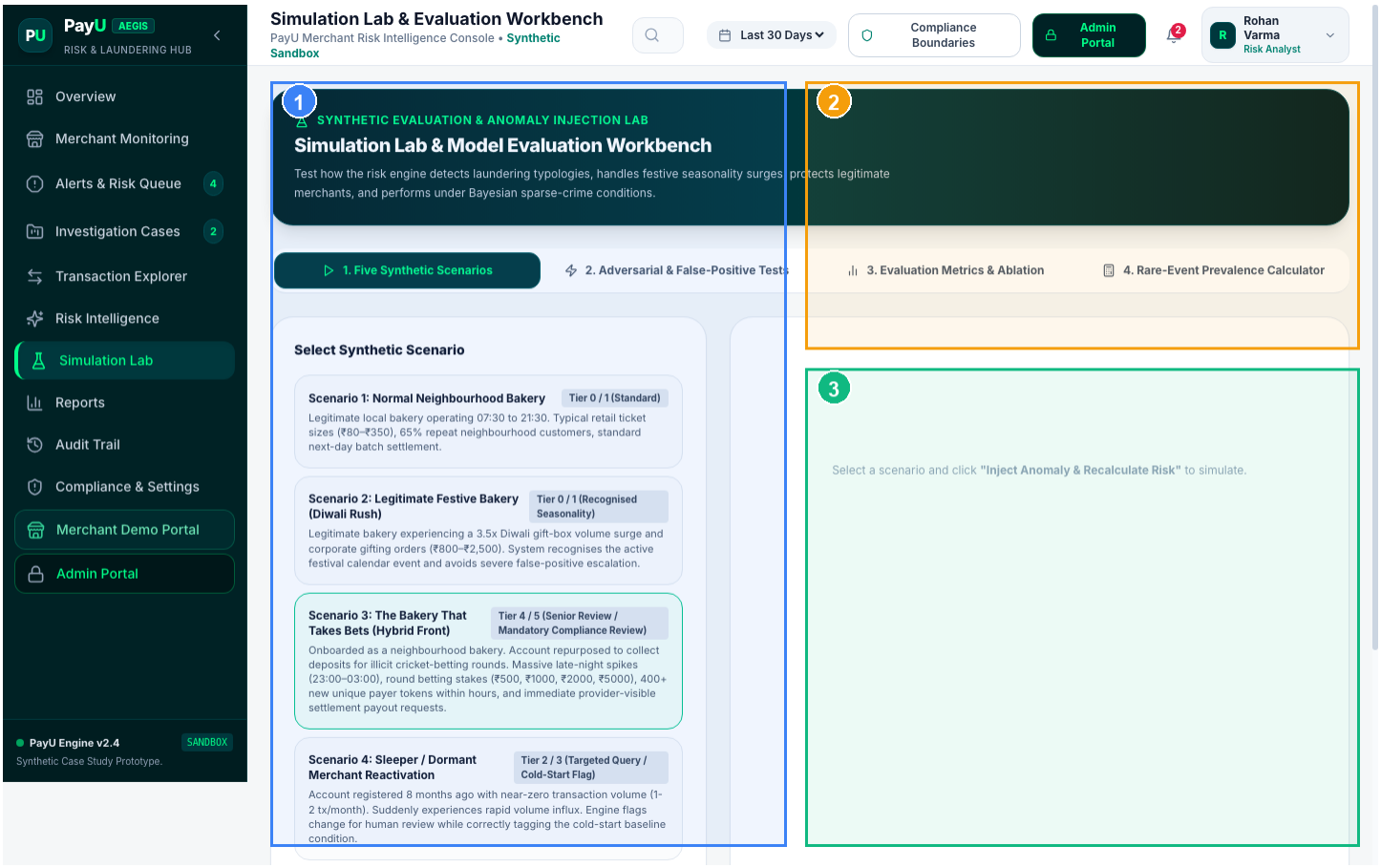
AEGIS-PAY does not rely on opaque black-box scoring. Every risk score is calculated from ten transparent, explainable behavioral signals. The Risk Intelligence view provides the mathematical definition, baseline formula, and weight contribution for each signal.



Mark	Feature Name	What It Means & How It Works	How To Use It
[1]	**Signal Category Filter Bar**	Groups the ten signals into four operational categories: Timing Anomaly, Amount Distribution, Payer Relationship, and Settlement Flow.	Click a category button to inspect related signals and compare baseline thresholds across different retail sectors.
[2]	**Signal Card & Weight Indicator**	Displays the signal code (e.g. SIG_TIMING_OFFHOURS), its human-readable title, its percentage weight (e.g. 15%), and its current activation state.	Review signal weights to understand which behaviors drive the highest risk tier escalations in the scoring engine.
[3]	**Mathematical Formula & Baseline**	Shows the mathematical equation used to trigger the signal. For example, ticket deviation calculates z-scores against category median values.	Use these formulas during merchant audits to explain to compliance reviewers why a merchant was flagged.
[4]	**Calibration Threshold Sliders**	Displays the trigger sensitivity thresholds (e.g. off-hours volume ratio exceeding 25% of daily total over a rolling 48-hour window).	Compliance officers can review sensitivity parameters to ensure thresholds remain calibrated to current seasonal conditions.

7. Simulation Lab and Scenario Stress Testing

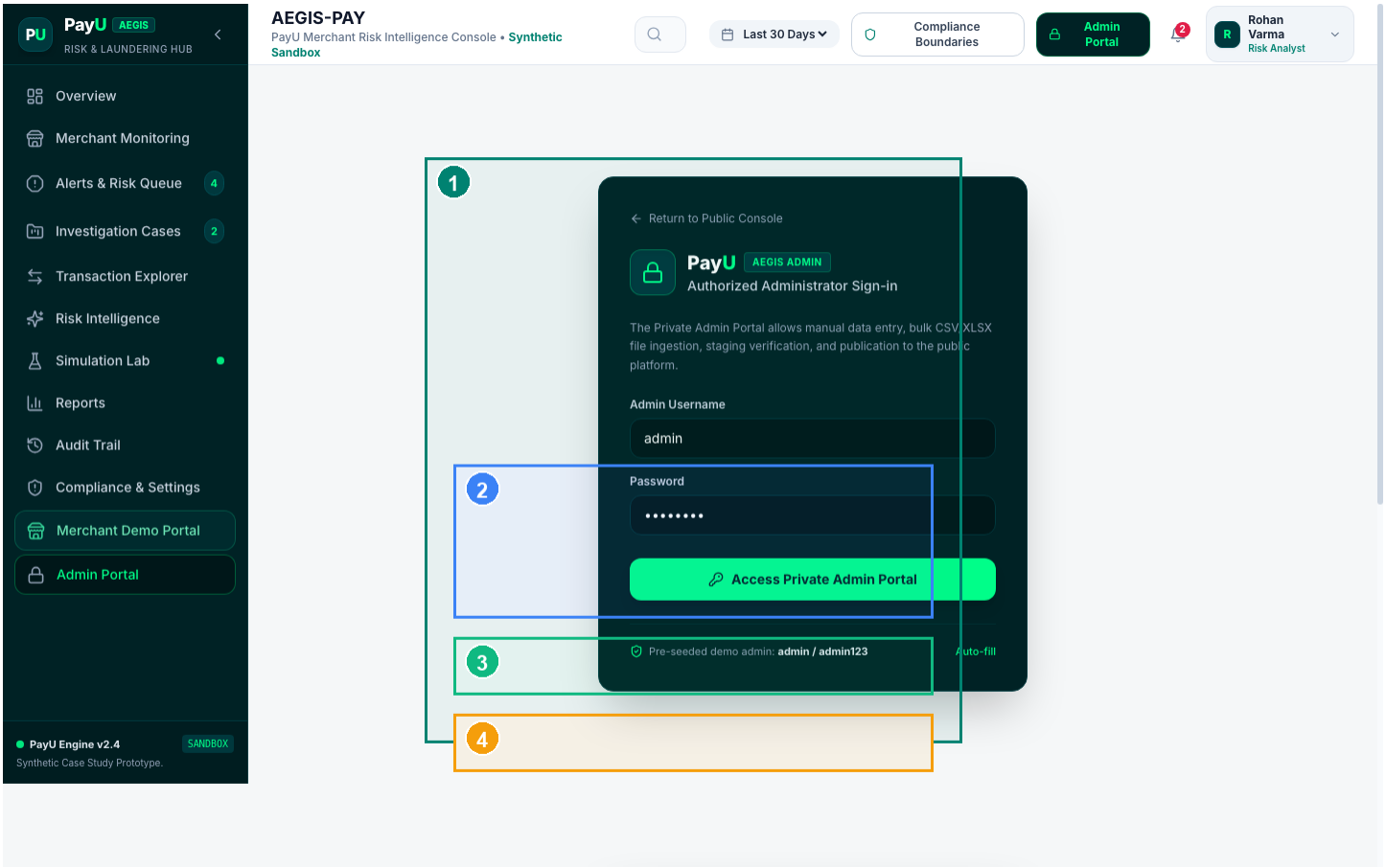
The Simulation Lab allows analysts to execute synthetic stress tests without affecting live production data. It includes 5 core scenarios and 12 adversarial false-positive tests, helping teams calibrate thresholds and distinguish legitimate festive volume spikes from actual transaction-laundering schemes.



Mark	Feature Name	What It Means & How It Works	How To Use It
[1]	**Scenario Selector & Typology List**	Displays pre-configured synthetic merchant scenarios: Scenario 1 Normal Bakery, Scenario 2 Legitimate Festive Diwali Bakery, Scenario 3 The Bakery That Takes Bets (Hybrid Front), Scenario 4 High-Velocity Layering, and Scenario 5 Dormant Reactivation.	Select any scenario card on the left to load its characteristics, expected risk tier, and synthetic transaction batch.
[2]	**Scenario Dossier & Behavioral Rules**	Details the target merchant ID, operating hours, expected ticket sizes (e.g. ■80 to ■350), customer repeat rate, and settlement speed.	Read the typology explanation to see why Festive Bakery (Scenario 2) stays in Tier 1 while Hybrid Front (Scenario 3) escalates to Tier 5.
[3]	**Simulation Runner & Impact Delta**	Contains the 'Run Simulation' trigger button, batch size override, and the Before vs After calibration impact gauge.	Click 'Execute Simulation' to run the synthetic batch through the scoring engine. Watch the risk score and tier adjust in real time.

8. Private Admin Portal and Data Ingestion

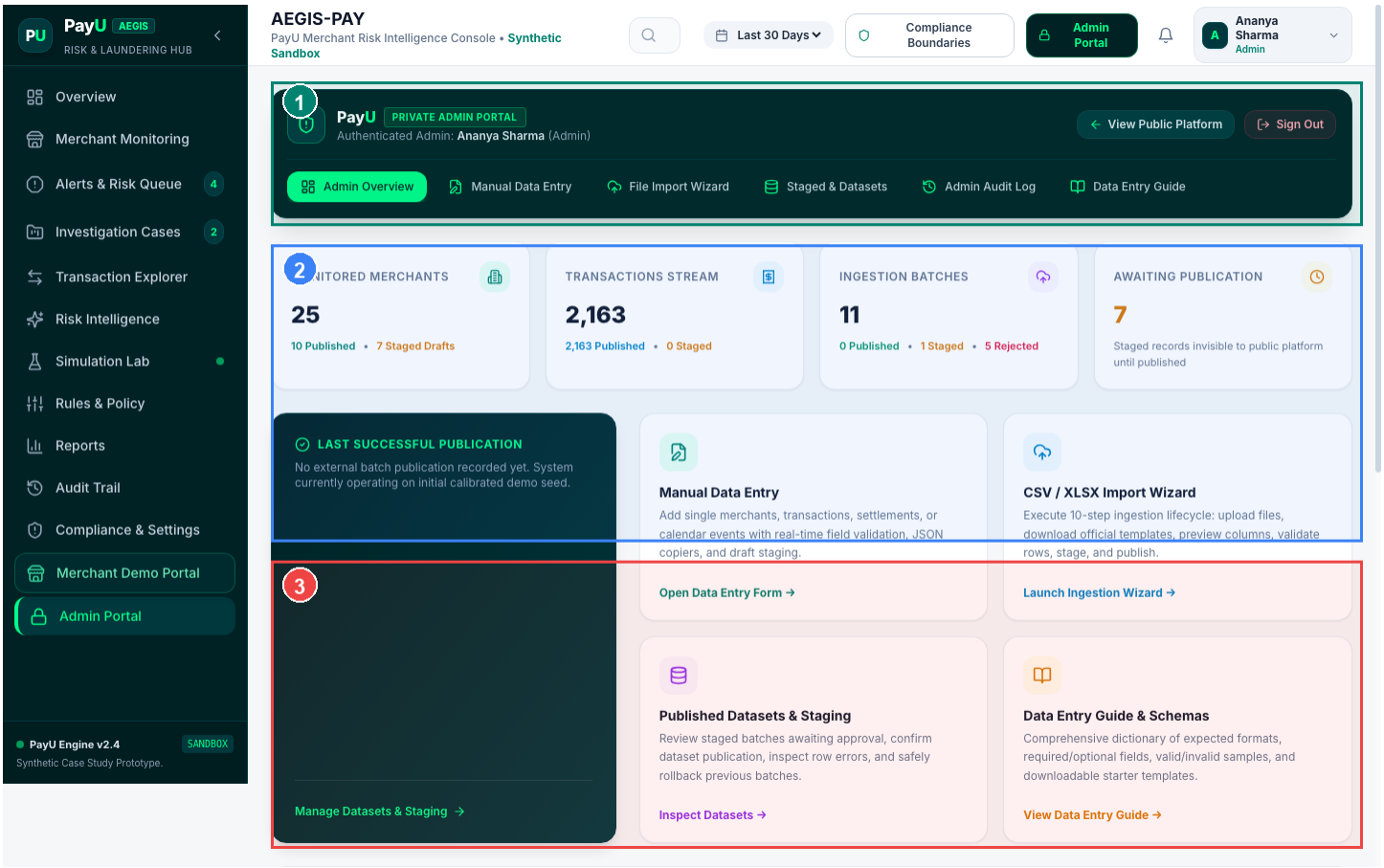
The Private Admin Portal gives platform administrators full control over dataset ingestion. It supports manual data entry with strict format validation, bulk CSV and Excel imports with staging verification, and one-click rollback to protect data integrity.



Mark	Feature Name	What It Means & How It Works	How To Use It
[1]	**Admin Security Gate & Header**	Protected entry point to the administrative ingestion engine. Non-admin users are restricted to public risk console views.	Navigate directly to /admin/login or click 'Admin Portal' in the top right header navigation to access this sign-in screen.
[2]	**Credential Input Fields**	Secure inputs for Admin Username and Password. Authenticates via cryptographic JWT token with role-based access validation.	Enter username 'admin' and password 'admin123' to authenticate into the administrative session.
[3]	**Access Private Admin Portal Button**	Submits the sign-in request and loads the administrative ingestion telemetry dashboard upon successful authentication.	Click the green button or press Enter to sign in. The navigation bar will automatically update with admin tools.
[4]	**Auto-Fill Demo Credentials**	A convenient shortcut button that instantly populates the username and password fields for testing and demonstrations.	Click 'Auto-fill' to load 'admin / admin123' in one click without typing, then click Access Private Admin Portal.

9. Admin Telemetry, Manual Entry & File Import Wizard

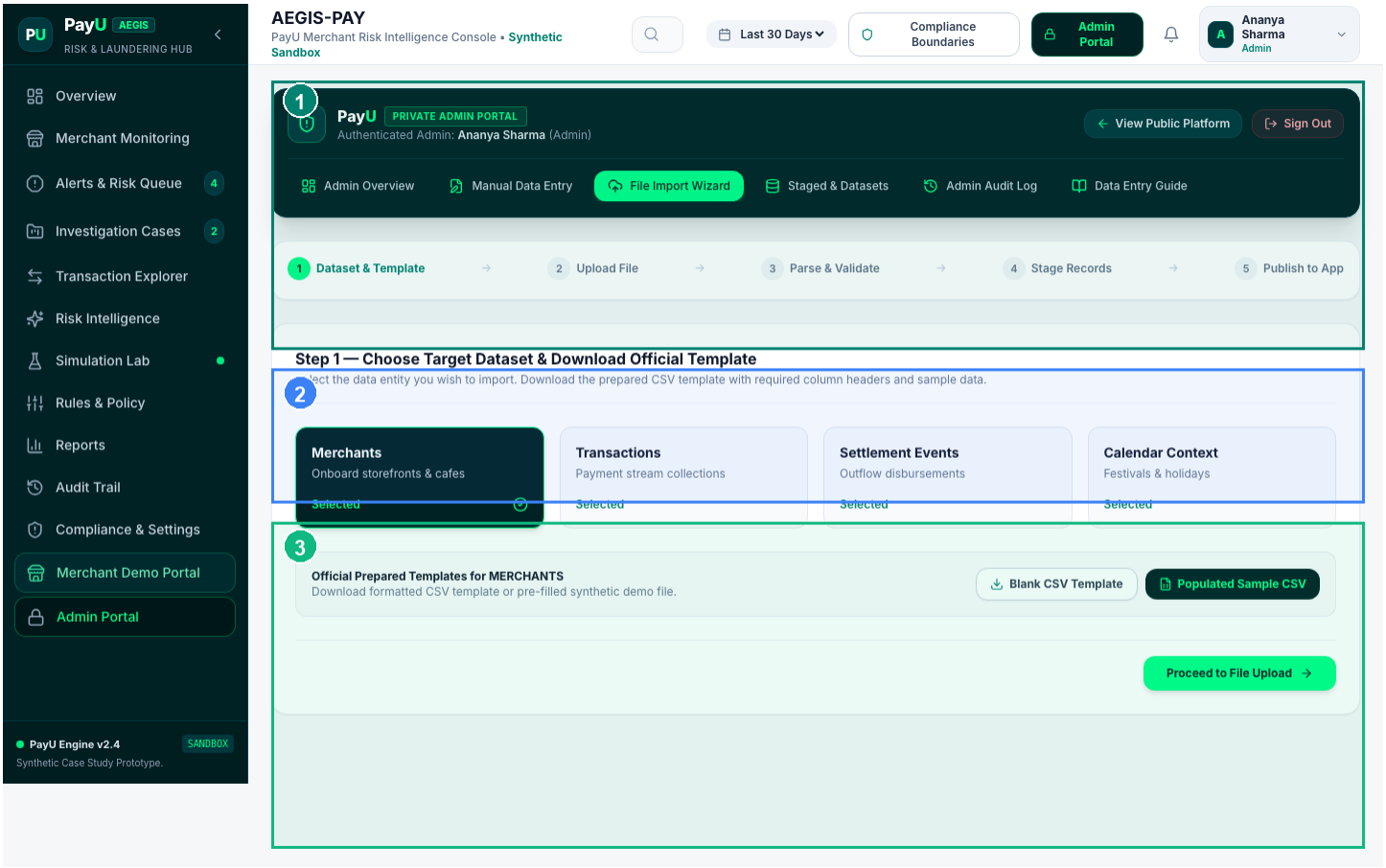
Inside the Admin Portal, administrators have access to three core operational workspaces: the Ingestion Overview (tracking published vs draft records), Manual Data Entry (strict field-by-field entry), and the Bulk File Import Wizard (automated multi-step CSV/Excel processing with staging and rollback).



Mark	Feature Name	What It Means & How It Works	How To Use It
[1]	**Ingestion Telemetry Counters**	Three key telemetry cards: Total Merchants (published vs staged drafts), Total Transactions (published vs draft), and Import Batches (staged, published, rejected).	Check 'Records Awaiting Publication' to see how many draft records are staged before approving them for live console visibility.
[2]	**Staged Batches & Review Pipeline**	Lists all recent file import batches with status: 'Staged', 'Published', or 'Rejected'. Displays row counts, error counts, and upload timestamps.	Click 'Review & Publish' on any staged batch to inspect parsed rows, verify validation status, and publish to the public console.
[3]	**Administrative Audit & Rollback Center**	A chronological audit stream of all manual creations, batch imports, publications, and rollbacks with actor names and rationales.	Click 'Rollback Batch' on any published dataset to instantly unpublish all associated records if an error is identified after launch.

10. Bulk Import Wizard and Manual Data Entry

AEGIS-PAY provides a 10-step wizard lifecycle for importing CSV and XLSX files. Files undergo strict header mapping, format validation, formula injection protection, and staging verification before publication.



Mark	Feature Name	What It Means & How It Works	How To Use It
[1]	**File Dropzone & Format Selector**	Supports drag-and-drop file upload for CSV and XLSX files. Select dataset type: Merchants, Transactions, Settlement Events, or Calendar Events.	Drag your CSV file into the dropzone or click to browse. Download starter templates using the 'Download Starter Template' button.
[2]	**10-Step Ingestion Lifecycle Wizard**	Step-by-step progress tracking: File Upload, Header Mapping, Format Validation, Formula Neutralization, Staging Preview, Approval, and Publication.	Watch the wizard validate rows automatically. If format errors are detected, click 'Download Error Report CSV' to see row-by-row details.
[3]	**Staging Review & One-Click Publish**	Displays a live preview table of all parsed rows with publication mode toggle (Staged Draft vs Direct Published).	Verify parsed columns and click 'Publish to Public Console'. Records instantly become visible to analysts across all monitoring views.

11. Operational Glossary and Troubleshooting FAQ

Reference definitions for core terms and troubleshooting solutions for common operating scenarios.

Term	Operational Definition & Compliance Meaning
Risk Score (0 - 1000)	A weighted composite score calculated from 10 behavioral signals. Higher scores reflect greater divergence from normal retail behavior.
Tier 0 (Standard)	Score 0 to 250. Normal operating retail profile. Standard next-day settlement and routine passive monitoring.
Tier 1 (Passive Monitoring)	Score 251 to 400. Mild seasonal volume variation (e.g. Diwali festive gift boxes). No merchant contact required.
Tier 2 (Targeted Query)	Score 401 to 600. Noticeable ticket deviation or reactivation after dormancy. Analyst dispatches clarification inquiry to merchant.
Tier 3 (Active Review)	Score 601 to 750. Significant off-hours sales spikes or round-amount clusters. Analyst initiates formal case dossier.
Tier 4 (Enhanced Due Diligence)	Score 751 to 900. High-velocity layering or repeat card dropping. Settlement hold applied pending senior compliance review.
Tier 5 (Decision Required)	Score 901 to 1000. Critical policy anomaly (e.g. hybrid gambling front). Mandatory senior compliance review and containment.
Pseudonymous Token	A privacy-preserving cryptographic hash (tok_...) replacing customer card numbers and bank accounts to eliminate PII risk.
Four-Eyes Review	Governance rule requiring at least two authorized individuals (Risk Analyst and Compliance Reviewer) before any containment action.

Frequently Asked Questions (FAQ)

Q: How do I access the web application?
A: Open your web browser and go to https://aegis-pay-beta.vercel.app. The platform is public and immediately accessible without any installation.

Q: How do I switch between analyst and admin personas?
A: In the top right header, click the user chip (Demo Analyst). A dropdown menu opens showing all available demo roles. Select 'Ananya Sharma (Admin)' to switch to Admin mode, or click 'Admin Portal' in the navigation bar.

Q: Why does the Festive Bakery not trigger an alert during Diwali?
A: The Festive Bakery (Scenario 2) experiences high sales volume, but its transactions occur during daytime business hours and reflect normal retail pricing. AEGIS-PAY compares volume against calendar holiday events and peer categories, preventing false positives.

Q: Can an ingested dataset be rolled back if an error was published?
A: Yes. In the Private Admin Portal, navigate to the Datasets tab. Locate the import batch and click 'Rollback'. All associated merchant and transaction records will be unpublished immediately with a mandatory audit rationale.